

Tencent Holdings Ltd ADR TCEHY ★★★★★ 1 Sep 2026 21:41, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation
56.09 USD 1 Sep 2026	99.00 USD 12 Aug 2026 23:08, UTC	0.57	506.88 USD Bil 2 Sep 2026	Wide	Large Blend	High	Exemplary

Ivan Su

Director

Morningstar

+852 2973 4609

ruzhe.su@morningstar.com

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The primary analyst covering this company does not own its stock.

Reporting Currency: CNY | Trading Currency: USD
Currency amounts expressed with "\$" are in U.S. dollars (USD) unless otherwise denoted.

Tencent: The AI Cash Drain Has a Visible End Date; HKD 780 Fair Value Unchanged

Analyst Note Ivan Su, Director, 2 Sep 2026

Recent Tencent commentary provides more visibility into when free cash flow turns positive, since the memory prepayments driving recent outflow are lump sum rather than recurring. It also offers early clues on consumer AI product economics that appear more favorable than we had assumed.

Why it matters: Tencent's artificial intelligence buildout won't be a perpetual cash burn, with downside limited if returns disappoint. Free cash flow should turn positive in 2027, opening room for larger buybacks from that point.

► Tencent is directing most compute to train its Hunyuan large language model. The model should therefore improve rapidly, and if it doesn't, the same capacity gets rented out to third parties, an immediate path to recouping the steep investment.

► That said, we still believe the largest long-term AI value accrues to the applications (advertising, games, and the WeChat AI agent), not to renting out compute, since these applications sit on network effects moats that a better model from a competitor cannot easily displace.

The bottom line: We maintain our HKD 780 fair value estimate on wide-moat Tencent. Shares trade at 15 times earnings and a 50% discount to our fair value, which we view as very undervalued.

► An equity raise is off the table. The company explicitly rejected one and will fund AI via operating cash flow and sales from its listed investment portfolio (USD 72 billion). This removes a dilution risk that had been on some investors' minds.

Financial Summary and Key Statistics

	Actual		Forecast	
	2024	2025	2026	2027
Revenue (CNY Mil)	660,257	751,766	833,359	922,235
Revenue Growth %	8.4	13.9	10.9	10.7
Operating Income (Mil)	200,097	244,739	255,656	239,236
Operating Margin %	30.3	32.6	30.7	25.9
Adjusted EBITDA (Mil)	293,675	341,498	402,530	418,311
Adjusted EBITDA Margin %	44.5	45.4	48.3	45.4
Earnings Per Share (Diluted) (CNY)	20.49	24.15	24.82	22.80
Adjusted Earnings Per Share (Diluted) (CNY)	23.67	28.09	29.85	27.82
Adjusted EPS Growth %	44.3	18.7	6.3	-6.8
Price/Earnings	16.4	19.0	12.7	13.6
Price/Book	3.8	4.3	2.7	2.5
EV/EBITDA	12.3	14.2	17.1	16.5
Free Cash Flow Yield %	—	—	—	—

Source: Morningstar Valuation Model. Data as of 13 Aug 2026.

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Sector	Industry
 Communication Services	Internet Content & Information

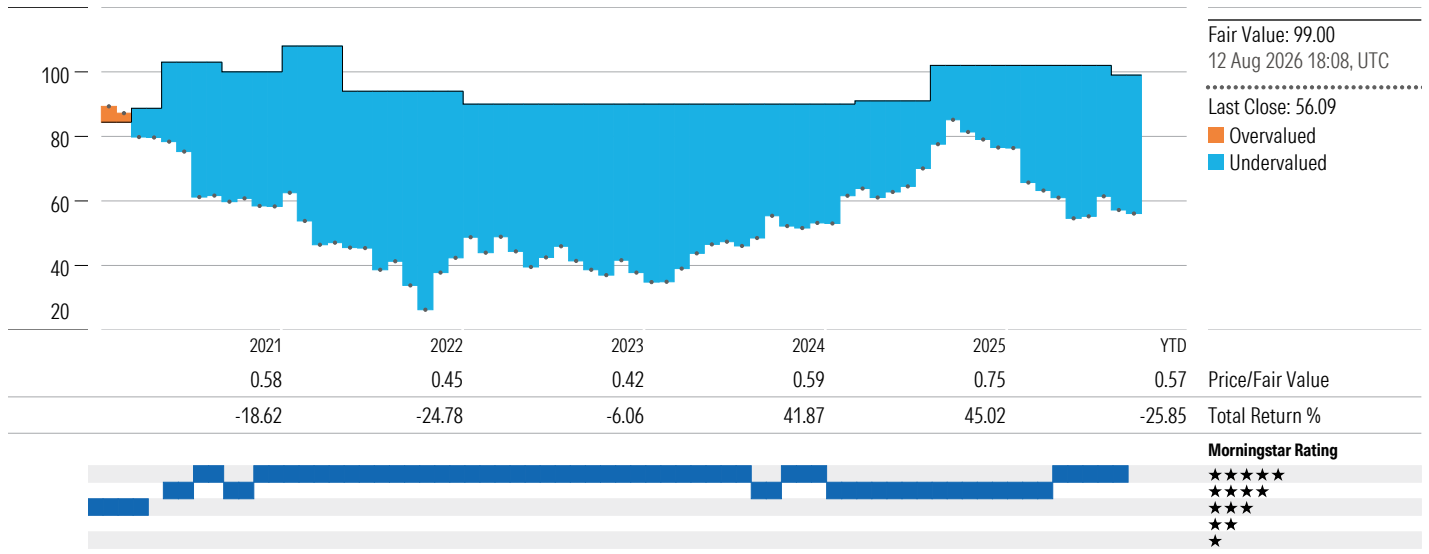
Business Description

Tencent is the world's largest game publisher, with top-grossing mobile hits like Honor of Kings and Peacekeeper Elite and a steady pipeline of new titles. It also operates WeChat—China's super-app with roughly 1.3 billion users—embedded in daily life for messaging, short video, mini programs, payments, and shopping. Beyond its own platforms, Tencent is a prolific strategic investor, holding stakes in leading internet companies such as PDD, Kuaishou, and Xiaohongshu.

- Sentiment should improve as Hunyuan's next-generation model advances in capability, and as the market recognizes that AI progress by competitors doesn't disrupt Tencent's WeChat and games moats.
- Between the lines:** Recent softness in advertising gross margin comes from deploying AI recommendations into short video content feeds, which improves the user experience and creates new inventory to monetize later. This supports our double-digit ad revenue growth call.
- There are two AI products we haven't yet baked into our valuation, but that increasingly look to carry upside. The first is the WeChat AI agent, which remains in pilot testing. The economics appear far more favorable than we previously assumed. The agent runs on a custom small AI model, and is said to cover 70% of average users' daily functionality needs. If these economics hold, payback on the WeChat AI agent could arrive faster than the hefty capex line suggests, though we have not built any of this upside into our fair value estimate.
- The second is WorkBuddy, an AI assistant (which the company calls a "harness") through which users can select from multiple AI models to run tasks. When DeepSeek—one of the models available—raised prices roughly 100%-200%, usage through WorkBuddy did not fall. That points to some price inelasticity in Chinese AI inference demand and gives us more confidence in the eventual return profile of this product line. That said, Chinese frontier model capability still trails the global leaders, compute remains constrained, and Tencent's business of selling model capacity to enterprises is not yet at scale, so we continue to treat WorkBuddy as potential upside not in our HKD 780 fair value estimate. ■■

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Price vs. Fair Value



Competitors

	Tencent Holdings Ltd ADR TCEHY	Baidu Inc 09888	NetEase Inc Ordinary Shares 09999	Alibaba Group ...rdinary Shares 09988
	Fair Value 99.00 Uncertainty: High Last Close 56.09	Fair Value 110.00 Uncertainty: High Last Close 93.70	Fair Value 309.00 Uncertainty: High Last Close 188.80	Fair Value 201.00 Uncertainty: High Last Close 110.40
Economic Moat	Wide	None	Narrow	Wide
Currency	USD	HKD	HKD	HKD
Fair Value	99.00 12 Aug 2026 23:08, UTC	110.00 1 Sep 2026 12:01, UTC	309.00 1 Sep 2026 12:12, UTC	201.00 1 Sep 2026 11:56, UTC
1-Star Price	153.45	170.50	478.95	311.55
5-Star Price	59.40	66.00	185.40	120.60
Assessment	Undervalued 1 Sep 2026	Undervalued 2 Sep 2026	Undervalued 2 Sep 2026	Undervalued 2 Sep 2026
Morningstar Rating	★★★★★ 1 Sep 2026 21:41, UTC	★★★★★ 1 Sep 2026 05:00, UTC	★★★★★ 1 Sep 2026 05:00, UTC	★★★★★ 1 Sep 2026 05:00, UTC
Analyst	Ivan Su, Director	Chelsey Tam, Senior Equity Analyst	Ivan Su, Director	Chelsey Tam, Senior Equity Analyst
Capital Allocation	Exemplary	Standard	Standard	Standard
Price/Fair Value	0.57	0.85	0.61	0.55
Price/Sales	4.58	1.81	4.68	1.82
Price/Book	3.00	0.81	3.13	1.75
Price/Earnings	15.43	—	16.79	27.39
Dividend Yield	1.20%	0.00%	2.50%	0.93%
Market Cap	—	254.25 Bil	604.40 Bil	2,195.30 Bil
52-Week Range	—	86.50—157.80	168.80—248.00	88.65—186.20
Investment Style	Large Blend	Large Value	Large Blend	Large Blend

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Morningstar Valuation Model Summary

Financials as of 13 Aug 2026

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Revenue (CNY Mil)	609,015	660,257	751,766	833,359	922,235	997,268	1,066,063	1,134,818
Operating Income (CNY Mil)	155,373	200,097	244,739	255,656	239,236	271,169	348,516	378,419
EBITDA (CNY Mil)	218,792	293,675	341,498	402,530	418,311	449,379	525,499	542,283
Adjusted EBITDA (CNY Mil)	218,792	293,675	341,498	402,530	418,311	449,379	525,499	542,283
Net Income (CNY Mil)	114,230	192,730	223,270	226,857	205,992	229,780	289,442	313,835
Adjusted Net Income (CNY Mil)	157,688	222,703	259,626	272,765	251,287	278,032	340,409	367,520
Free Cash Flow To The Firm (CNY Mil)	158,048	153,179	147,932	56,021	94,870	171,634	275,079	294,215
Weighted Average Diluted Shares Outstanding (Mil)	9,610	9,408	9,244	9,139	9,033	8,923	8,808	8,688
Earnings Per Share (Diluted) (CNY)	11.89	20.49	24.15	24.82	22.80	25.75	32.86	36.12
Adjusted Earnings Per Share (Diluted) (CNY)	16.41	23.67	28.09	29.85	27.82	31.16	38.65	42.30
Dividends Per Share (CNY)	3.07	4.15	4.89	4.96	4.56	6.44	9.86	10.84

Margins & Returns as of 13 Aug 2026

	3 Year Avg	Actual			Forecast					5 Year Avg
		2023	2024	2025	2026	2027	2028	2029	2030	
Operating Margin %	33.0	25.5	30.3	32.6	30.7	25.9	27.2	32.7	33.4	32.4
EBITDA Margin %	—	35.9	44.5	45.4	48.3	45.4	45.1	49.3	47.8	—
Adjusted EBITDA Margin %	—	35.9	44.5	45.4	48.3	45.4	45.1	49.3	47.8	47.2
Net Margin %	25.9	18.8	29.2	29.7	27.2	22.3	23.0	27.2	27.7	25.5
Adjusted Net Margin %	31.4	25.9	33.7	34.5	32.7	27.3	27.9	31.9	32.4	30.4
Free Cash Flow To The Firm Margin %	22.9	26.0	23.2	19.7	6.7	10.3	17.2	25.8	25.9	17.2

Growth & Ratios as of 13 Aug 2026

	3 Year CAGR	Actual			Forecast					5 Year CAGR
		2023	2024	2025	2026	2027	2028	2029	2030	
Revenue Growth %	10.7	9.8	8.4	13.9	10.9	10.7	8.1	6.9	6.5	8.6
Operating Income Growth %	33.5	51.1	28.8	22.3	4.5	-6.4	13.4	28.5	8.6	9.1
EBITDA Growth %	10.3	-19.6	34.2	16.3	17.9	3.9	7.4	16.9	3.2	9.9
Adjusted EBITDA Growth %	7.9	-19.6	34.2	16.3	17.9	3.9	7.4	16.9	3.2	9.7
Earnings Per Share Growth %	7.7	-38.5	72.3	17.9	2.8	-8.1	12.9	27.6	9.9	8.4
Adjusted Earnings Per Share Growth %	7.7	37.6	44.3	18.7	6.3	-6.8	12.0	24.0	9.5	8.4

Valuation as of 13 Aug 2026

	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Price/Earnings	16.3	16.4	19.0	12.7	13.6	12.1	9.8	9.0
Price/Sales	4.1	5.5	6.5	8.3	7.5	6.9	6.5	6.1
Price/Book	3.2	3.8	4.3	2.7	2.5	2.3	2.1	1.9
Price/Cash Flow	—	—	—	—	—	—	—	—
EV/EBITDA	11.4	12.3	14.2	17.1	16.5	15.3	13.1	12.7
EV/EBIT	16.1	18.0	19.8	27.0	28.8	25.4	19.8	18.2
Dividend Yield %	1.2	1.1	0.9	1.3	1.2	1.7	2.6	2.9
Dividend Payout %	18.7	17.5	17.4	16.6	16.4	20.7	25.5	25.6
Free Cash Flow Yield %	—	—	—	—	—	—	—	—

Operating Performance / Profitability as of 13 Aug 2026

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
ROA %	7.2	10.8	11.0	10.2	8.6	9.1	10.8	11.0
ROE %	13.1	18.3	18.0	16.5	13.8	14.3	16.6	16.7
ROIC %	154.1	92.1	86.4	60.9	40.9	40.3	50.1	54.8

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Financial Leverage (Reporting Currency)

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Debt/Capital %	4.6	6.9	10.1	5.6	5.2	4.8	4.5	4.2
Assets/Equity	1.8	1.7	1.6	1.6	1.6	1.6	1.5	1.5
Net Debt/EBITDA	0.2	0.2	0.2	-0.2	-0.2	-0.4	-0.6	-0.8
Total Debt/EBITDA	1.2	1.1	1.1	0.9	0.9	0.8	0.7	0.7
EBITDA/ Net Interest Expense	-142.1	-73.0	-192.0	-90.1	-197.9	-303.3	-199.3	-112.3

Forecast Revisions as of 12 Aug 2026

Prior data as of 11 Aug 2026	2026		2027		2028	
	Current	Prior	Current	Prior	Current	Prior
Fair Value Estimate Change (Trading Currency)	99.00	800.88	—	—	—	—
Revenue (CNY Mil)	833,359	825,326	922,235	906,284	997,268	979,302
Operating Income (CNY Mil)	255,656	264,379	239,236	309,174	271,169	340,715
EBITDA (CNY Mil)	402,530	379,787	418,311	433,603	449,379	473,273
Net Income (CNY Mil)	272,765	284,623	251,287	307,989	278,032	335,521
Earnings Per Share (Diluted) (CNY)	24.82	26.42	22.80	29.00	25.75	32.24
Adjusted Earnings Per Share (Diluted) (CNY)	29.85	31.30	27.82	34.42	31.16	38.08
Dividends Per Share (CNY)	4.96	5.28	4.56	5.80	6.44	8.06

Key Valuation Drivers as of 13 Aug 2026

Cost of Equity %	9.8
Pre-Tax Cost of Debt %	5.3
Weighted Average Cost of Capital %	8.6
Long-Run Tax Rate %	—
Stage II EBI Growth Rate %	4.0
Stage II Investment Rate %	12.5
Perpetuity Year	20

Additional estimates and scenarios available for download at <https://pitchbook.com/>.

Discounted Cash Flow Valuation as of 13 Aug 2026

	CNY Mil
Present Value Stage I	1,639,454
Present Value Stage II	1,455,907
Present Value Stage III	1,639,249
Total Firm Value	4,734,611
Cash and Equivalents	455,121
Debt	377,191
Other Adjustments	1,065,015
Equity Value	5,877,556
Projected Diluted Shares	9,139
Fair Value per Share (USD)	99.00

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Income Statement (CNY)

Fiscal Year, ends 31 Dec

	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Revenue (Mil)	609,015	660,257	751,766	833,359	922,235	997,268	1,066,063	1,134,818
Cost of Goods Sold (Mil)	315,906	311,011	329,173	362,149	415,551	433,899	435,040	459,077
Gross Profit (Mil)	293,109	349,246	422,593	471,210	506,684	563,368	631,022	675,741
Selling, General, Administrative & Other Expenses (Mil)	39,447	42,075	50,380	49,157	61,790	64,822	67,162	70,359
Advertising & Marketing Expenses (Mil)	34,211	36,388	41,727	49,576	48,878	52,855	55,435	68,089
Research & Development (Mil)	64,078	70,686	85,747	116,822	156,780	174,522	159,909	158,875
Depreciation & Amortization (if reported separately) (Mil)	0	0	0	0	0	0	0	0
Adjusted Operating Income (Mil)	155,373	200,097	244,739	255,656	239,236	271,169	348,516	378,419
Financial Non-Cash (Gains)/Losses (Mil)	-4,411	-37,365	-30,731	-21,112	-25,000	-25,000	-25,000	-25,000
Irregular Cash (Gains)/Losses (Mil)	0	0	0	0	0	0	0	0
Operating Income (Mil)	159,784	237,462	275,470	276,768	264,236	296,169	373,516	403,419
Net Interest Expense (Mil)	-1,540	-4,023	-1,779	-4,469	-2,114	-1,482	-2,636	-4,830
Income Tax Expense (Mil)	43,276	45,018	47,448	47,978	56,661	63,860	81,915	89,297
After-Tax Items (Mil)	-986	-1,343	-1,572	-1,234	-1,234	-1,234	-1,234	-1,234
(Minority Interest) (Mil)	-2,832	-2,394	-4,959	-5,168	-2,464	-2,777	-3,562	-3,882
Net Income (Mil)	114,230	192,730	223,270	226,857	205,992	229,780	289,442	313,835
Adjusted Net Income (Mil)	157,688	222,703	259,626	272,765	251,287	278,032	340,409	367,520
Weighted Average Diluted Shares Outstanding (Mil)	9,610	9,408	9,244	9,139	9,033	8,923	8,808	8,688
Diluted Earnings Per Share	11.89	20.49	24.15	24.82	22.80	25.75	32.86	36.12
Diluted Adjusted Earnings Per Share	16.41	23.67	28.09	29.85	27.82	31.16	38.65	42.30
Dividends Per Common Share (CNY)	3.07	4.15	4.89	4.96	4.56	6.44	9.86	10.84
EBITDA (Mil)	218,792	293,675	341,498	402,530	418,311	449,379	525,499	542,283
Adjusted EBITDA (Mil)	218,792	293,675	341,498	402,530	418,311	449,379	525,499	542,283

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Key Cash Flow Items (CNY)

Fiscal Year, ends 31 Dec	Actual			Forecast as of 13 Aug				
	2023	2024	2025	2026	2027	2028	2029	2030
Cash from Working Capital (Mil)	24,552	9,343	15,967	6,321	26,133	726	5,992	13,453
(Capital Expenditures) (Mil)	-53,445	-95,487	-119,512	-278,100	-264,666	-185,919	-145,276	-143,164
Depreciation (Mil)	26,305	27,332	32,799	86,135	117,134	118,128	118,085	105,588
Amortization (Mil)	32,703	28,881	33,229	39,627	36,941	35,082	33,898	33,276
Net New (Investment), Organic (Mil)	30,115	-29,931	-37,517	-146,017	-84,457	-31,982	12,700	9,153
(Purchases)/Sales of Companies & Assets (Mil)	-8,914	-4,942	-34,620	0	0	0	0	0
Net New (Investment), Total (Mil)	21,201	-34,873	-72,137	-146,017	-84,457	-31,982	12,700	9,153
Other Non-Cash Items, From Cash Flows (Mil)	28,155	35,960	29,005	0	0	0	0	0
Free Cash Flow to the Firm (Mil)	158,048	153,179	147,932	56,021	94,870	171,634	275,079	294,215

Balance Sheet (CNY)

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Assets								
Cash and Equivalents (Bil)	391	406	455	443	470	548	690	833
Inventory (Bil)	2	0	0	1	1	1	1	1
Accounts Receivable (Bil)	47	48	50	64	71	77	82	87
Net Property, Plant and Equipment (Bil)	53	80	150	317	438	478	475	481
Goodwill (Bil)	0	0	0	0	0	0	0	0
Other Intangibles (Bil)	178	196	206	190	179	172	167	165
Other Operating Assets (Bil)	198	226	215	238	263	285	304	324
Non-Operating Assets (Bil)	710	824	963	963	963	963	963	963
Total Assets (Bil)	1,579	1,781	2,039	2,216	2,385	2,522	2,682	2,854
Liabilities								
Accounts Payable (Bil)	101	119	121	129	148	143	143	151
Debt (Bil)	334	330	377	377	377	377	377	377
Other Operating Liabilities (Bil)	268	278	300	331	365	394	421	447
Non-Operating Liabilities (Bil)	0	0	0	0	0	0	0	0
Total Liabilities (Bil)	704	727	798	837	890	914	941	975
Equity								
Shareholders' Equity (Bil)	809	974	1,154	1,282	1,388	1,493	1,618	1,748
Minority Interest (Bil)	65	80	87	96	107	115	123	131
Total Equity (Bil)	874	1,054	1,241	1,379	1,495	1,608	1,741	1,879

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ESG Risk Rating Breakdown

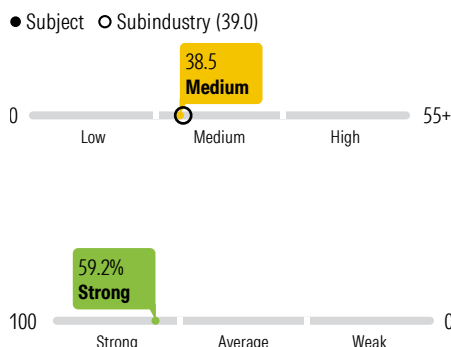
Exposure

Company Exposure ¹	38.5
- Manageable Risk	35.7
Unmanageable Risk²	2.7

Management

Manageable Risk	35.7
- Managed Risk ³	21.1
Management Gap⁴	14.6

Overall Unmanaged Risk 17.3



- Exposure represents a company's vulnerability to ESG risks driven by their business model
- Exposure is assessed at the Subindustry level and then specified at the company level
- Scoring ranges from 0-55+ with categories of low, medium, and high-risk exposure
- Management measures a company's ability to manage ESG risks through its commitments and actions
- Management assesses a company's efficiency on ESG programs, practices, and policies
- Management score ranges from 0-100% showing how much manageable risk a company is managing

ESG Risk Rating



ESG Risk Rating is of Sep 02, 2026. Highest Controversy Level is as of Aug 08, 2026. Sustainability Subindustry: Internet Software and Services. Sustainability provides Morningstar with company ESG ratings and metrics on a monthly basis and as such, the ratings in Morningstar may not necessarily reflect current Sustainability's scores for the company. For the most up to date rating and more information, please visit: sustainability.com/esg-ratings/.

ESG Risk Ratings measure the degree to which a company's value is impacted by environmental, social, and governance risks, by evaluating the company's ability to manage the ESG risks it faces.

1. A company's Exposure to material ESG issues 2. Unmanageable Risk refers to risks that are inherent to a particular business model that cannot be managed by programs or initiatives 3. Managed Risk = Manageable Risk multiplied by a Management score of 59.2% 4. Management Gap assesses risks that are not managed, but are considered manageable

Peer Analysis 02 Sep 2026

Peers are selected from the company's Sustainability-defined Subindustry and are displayed based on the closest market cap values

Company Name	Exposure	Management	ESG Risk Rating
Tencent Holdings Ltd	38.5 Medium 0 —●— 55+	59.2 Strong 100 —●— 0	17.3 Low 0 —●— 40+
Baidu Inc	40.8 Medium 0 —●— 55+	60.0 Strong 100 —●— 0	18.1 Low 0 —●— 40+
NetEase Inc	24.5 Low 0 —●— 55+	45.7 Average 100 —●— 0	13.9 Low 0 —●— 40+
Alibaba Group Holding Ltd	37.6 Medium 0 —●— 55+	62.1 Strong 100 —●— 0	16.0 Low 0 —●— 40+
Meta Platforms Inc	44.3 Medium 0 —●— 55+	49.1 Average 100 —●— 0	24.4 Medium 0 —●— 40+

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Research Methodology for Valuing Companies

Overview

At the heart of our valuation system is a detailed projection of a company's future cash flows, resulting from our analysts' research. Analysts create custom industry and company assumptions to feed income statement, balance sheet, and capital investment assumptions into our globally standardized, proprietary discounted cash flow, or DCF, modeling templates. We use scenario analysis, in-depth competitive advantage analysis, and a variety of other analytical tools to augment this process. Moreover, we think analyzing valuation through discounted cash flows presents a better lens for viewing cyclical companies, high-growth firms, businesses with finite lives (e.g., mines), or companies expected to generate negative earnings over the next few years. That said, we don't dismiss multiples altogether but rather use them as supporting cross-checks for our DCF-based fair value estimates. We also acknowledge that DCF models offer their own challenges (including a potential proliferation of estimated inputs and the possibility that the method may miss short-term market-price movements), but we believe these negatives are mitigated by deep analysis and our long-term approach.

Morningstar's equity research group ("we," "our") believes that a company's intrinsic worth results from the future cash flows it can generate. The Morningstar Rating for stocks identifies stocks trading at a discount or premium to their intrinsic worth—or fair value estimate, in Morningstar terminology. Five-star stocks sell for the biggest risk adjusted discount to their fair values, whereas 1-star stocks trade at premiums to their intrinsic worth.

Four key components drive the Morningstar rating: (1) our assessment of the firm's economic moat, (2) our estimate of the stock's fair value, (3) our uncertainty around that fair value estimate and (4) the current market price. This process ultimately culminates in our singlepoint star rating.

1. Economic Moat

The concept of an economic moat plays a vital role not only in our qualitative assessment of a firm's long-term investment potential, but also in the actual calculation of our fair value estimates. An economic moat is a structural

feature that allows a firm to sustain excess profits over a long period of time. We define economic profits as returns on invested capital (or ROIC) over and above our estimate of a firm's cost of capital, or weighted average cost of capital (or WACC). Without a moat, profits are more susceptible to competition. We have identified five sources of economic moats: intangible assets, switching costs, network effect, cost advantage, and efficient scale.

Companies with a narrow moat are those we believe are more likely than not to achieve normalized excess returns for at least the next 10 years. Wide-moat companies are those in which we have very high confidence that excess returns will remain for 10 years, with excess returns more likely than not to remain for at least 20 years. The longer a firm generates economic profits, the higher its intrinsic value. We believe low-quality, no-moat companies will see their normalized returns gravitate toward the firm's cost of capital more quickly than companies with moats.

When considering a company's moat, we also assess whether there is a substantial threat of value destruction, stemming from risks related to ESG, industry disruption, financial health, or other idiosyncratic issues. In this context, a risk is considered potentially value destructive if its occurrence would eliminate a firm's economic profit on a cumulative or midcycle basis. If we deem the probability of occurrence sufficiently high, we would not characterize the company as possessing an economic moat.

2. Estimated Fair Value

Combining our analysts' financial forecasts with the firm's economic moat helps us assess how long returns on invested capital are likely to exceed the firm's cost of capital. Returns of firms with a wide economic moat rating are assumed to fade to the perpetuity period over a longer period of time than the returns of narrow-moat firms, and both will fade slower than no-moat firms, increasing our estimate of their intrinsic value.

Our model is divided into three distinct stages:

Stage I: Explicit Forecast

In this stage, which can last five to 10 years, analysts make full financial statement forecasts, including items such as revenue, profit margins, tax rates, changes in workingcapital accounts, and capital spending. Based on

these projections, we calculate earnings before interest, after taxes (EBI) and the net new investment (NNI) to derive our annual free cash flow forecast.

Stage II: Fade

The second stage of our model is the period it will take the company's return on new invested capital—the return on capital of the next dollar invested ("RONIC")—to decline (or rise) to its cost of capital. During the Stage II period, we use a formula to approximate cash flows in lieu of explicitly modeling the income statement, balance sheet, and cash flow statement as we do in Stage I. The length of the second stage depends on the strength of the company's economic moat. We forecast this period to last anywhere from one year (for companies with no economic moat) to 10–15 years or more (for wide-moat companies). During this period, cash flows are forecast using four assumptions: an average growth rate for EBI over the period, a normalized investment rate, average return on new invested capital (RONIC), and the number of years until perpetuity, when excess returns cease. The investment rate and return on new invested capital decline until a perpetuity value is calculated. In the case of firms that do not earn their cost of capital, we assume marginal ROICs rise to the firm's cost of capital (usually attributable to less reinvestment), and we may truncate the second stage.

Stage III: Perpetuity

Once a company's marginal ROIC hits its cost of capital, we calculate a continuing value, using a standard perpetuity formula. At perpetuity, we assume that any growth or decline or investment in the business neither creates nor destroys value and that any new investment provides a return in line with estimated WACC.

Because a dollar earned today is worth more than a dollar earned tomorrow, we discount our projections of cash flows in stages I, II, and III to arrive at a total present value of expected future cash flows. Because we are modeling free cash flow to the firm—representing cash available to provide a return to all capital providers—we discount future cash flows using the WACC, which is a weighted average of the costs of equity, debt, and preferred stock (and any other funding sources), using expected future proportionate long-term, market-value weights.

3. Uncertainty Around That Fair Value Estimate

Morningstar's Uncertainty Rating is designed to capture the range of potential outcomes for a company's intrinsic value. This rating is used to assign the margin of safety required before investing, which in turn explicitly drives our stock star rating system. The Uncertainty Rating is aimed at identifying the confidence we should have in as-

Morningstar Equity Research Star Rating Methodology



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signing a fair value estimate for a given stock.

Our Uncertainty Rating is meant to take into account anything that can increase the potential dispersion of future outcomes for the intrinsic value of a company, and anything that can affect our ability to accurately predict these outcomes. The rating begins with a suggested rating produced by a quantitative process based on the trailing 12-month standard deviation of daily stock returns. An analyst overlay is then applied, with analysts using the suggested rating, historical rating data, and their own knowledge of the company to inform them as they make the final Uncertainty Rating decision. Ultimately, the rating decision rests with the analyst. Analysts take into account many characteristics when making their final decision, including cyclical factors, operational and financial factors such as leverage, company-specific events, ESG risks, and anything else that might increase the potential dispersion of future outcomes and our ability to estimate those outcomes.

Our recommended margin of safety—the discount to fair value demanded before we'd recommend buying or selling the stock—widens as our uncertainty of the estimated value of the equity increases. The more uncertain we are about the potential dispersion of outcomes, the greater the discount we require relative to our estimate of the value of the firm before we would recommend the purchase of the shares. In addition, the Uncertainty Rating provides guidance in portfolio construction based on risk tolerance.

Our Uncertainty Ratings are: Low, Medium, High, Very High, and Extreme.

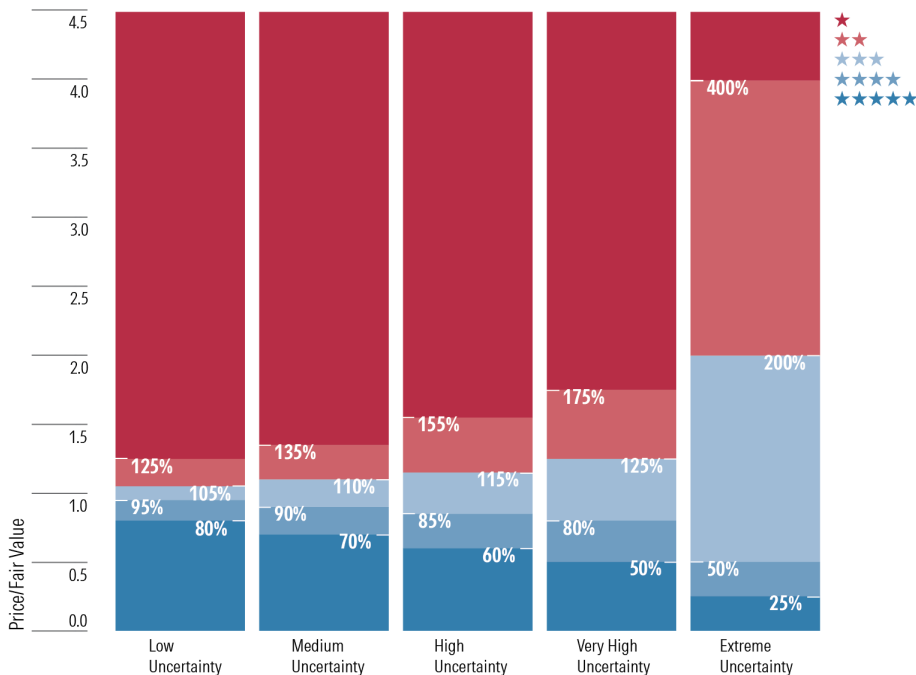
Margin of Safety		
Qualitative Analysis	★★★★★ Rating	★ Rating
Uncertainty Ratings		
Low	20% Discount	25% Premium
Medium	30% Discount	35% Premium
High	40% Discount	55% Premium
Very High	50% Discount	75% Premium
Extreme	75% Discount	300% Premium

Our uncertainty rating is based on the interquartile range, or the middle 50% of potential outcomes, covering the 25th percentile–75th percentile. This means that when a stock hits 5 stars, we expect there is a 75% chance that the intrinsic value of that stock lies above the current market price. Similarly, when a stock hits 1 star, we expect there is a 75% chance that the intrinsic value of that stock lies below the current market price.

4. Market Price

The market prices used in this analysis and noted in the report come from exchange on which the stock is listed which we believe is a reliable source.

Morningstar Equity Research Star Rating Methodology



For more details about our methodology, please go to <https://shareholders.morningstar.com>

Morningstar Star Rating for Stocks

Once we determine the fair value estimate of a stock, we compare it with the stock's current market price on a daily basis, and the star rating is automatically re-calculated at the market close on every day the market on which the stock is listed is open. Our analysts keep close tabs on the companies they follow, and, based on thorough and ongoing analysis, raise or lower their fair value estimates as warranted.

Please note, there is no predefined distribution of stars. That is, the percentage of stocks that earn 5 stars can fluctuate daily, so the star ratings, in the aggregate, can serve as a gauge of the broader market's valuation. When there are many 5-star stocks, the stock market as a whole is more undervalued, in our opinion, than when very few companies garner our highest rating.

We expect that if our base-case assumptions are true the market price will converge on our fair value estimate over time generally within three years (although it is impossible to predict the exact time frame in which market prices may adjust).

Our star ratings are guideposts to a broad audience and individuals must consider their own specific investment goals, risk tolerance, tax situation, time horizon, income needs, and complete investment portfolio, among other

factors.

The Morningstar Star Ratings for stocks are defined below:

★★★★★ We believe appreciation beyond a fair risk-adjusted return is highly likely over a multiyear time frame. Scenario analysis developed by our analysts indicates that the current market price represents an excessively pessimistic outlook, limiting downside risk and maximizing upside potential.

★★★★ We believe appreciation beyond a fair risk-adjusted return is likely.

★★★ Indicates our belief that investors are likely to receive a fair risk-adjusted return (approximately cost of equity).

★★ We believe investors are likely to receive a less than fair risk-adjusted return.

★ Indicates a high probability of undesirable risk-adjusted returns from the current market price over a multi-year time frame, based on our analysis. Scenario analysis by our analysts indicates that the market is pricing in an excessively optimistic outlook, limiting upside potential and leaving the investor exposed to Capital loss.

Other Definitions

Last Price: Price of the stock as of the close of the market of the last trading day before date of the report.

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Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments, and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

Capital Allocation (or Stewardship) analysis published prior to Dec. 9, 2020, was determined using a different process. Beyond investment strategy, financial leverage, and dividend and share buyback policies, analysts also considered execution, compensation, related party transactions, and accounting practices in the rating.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments, and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

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Sustainalytics ESG Risk Rating Assessment: The ESG Risk Rating Assessment is provided by Sustainalytics; a Morningstar company.

Sustainalytics' ESG Risk Ratings measure the degree to which company's economic value at risk is driven by environment, social and governance (ESG) factors.

Sustainalytics analyzes over 1,300 data points to assess a company's exposure to and management of ESG risks. In other words, ESG Risk Ratings measures a company's unmanaged ESG Risks represented as a quantitative score. Unmanaged Risk is measured on an open-ended scale starting at zero (no risk) with lower scores representing less unmanaged risk and, for 95% of cases, the unmanaged ESG Risk score is below 50.

Based on their quantitative scores, companies are grouped into one of five Risk Categories (negligible, low, medium, high, severe). These risk categories are absolute, meaning that a 'high risk' assessment reflects a comparable degree of unmanaged ESG risk across all subindustries covered.

The ESG Risk Rating Assessment is a visual representation of Sustainalytics ESG Risk Categories on a 1 to 5 scale. Companies with Negligible Risk = 5 Globes, Low Risk = 4, Medium Risk = 3 Globes, High Risk = 2 Globes, Severe Risk = 1 Globe. For more information, please visit sustainalytics.com/esg-ratings/

Ratings should not be used as the sole basis in evaluating a company or security. Ratings involve unknown risks and uncertainties which may cause our expectations not to occur or to differ significantly from what was expected and should not be considered an offer or solicitation to buy or sell a security.

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